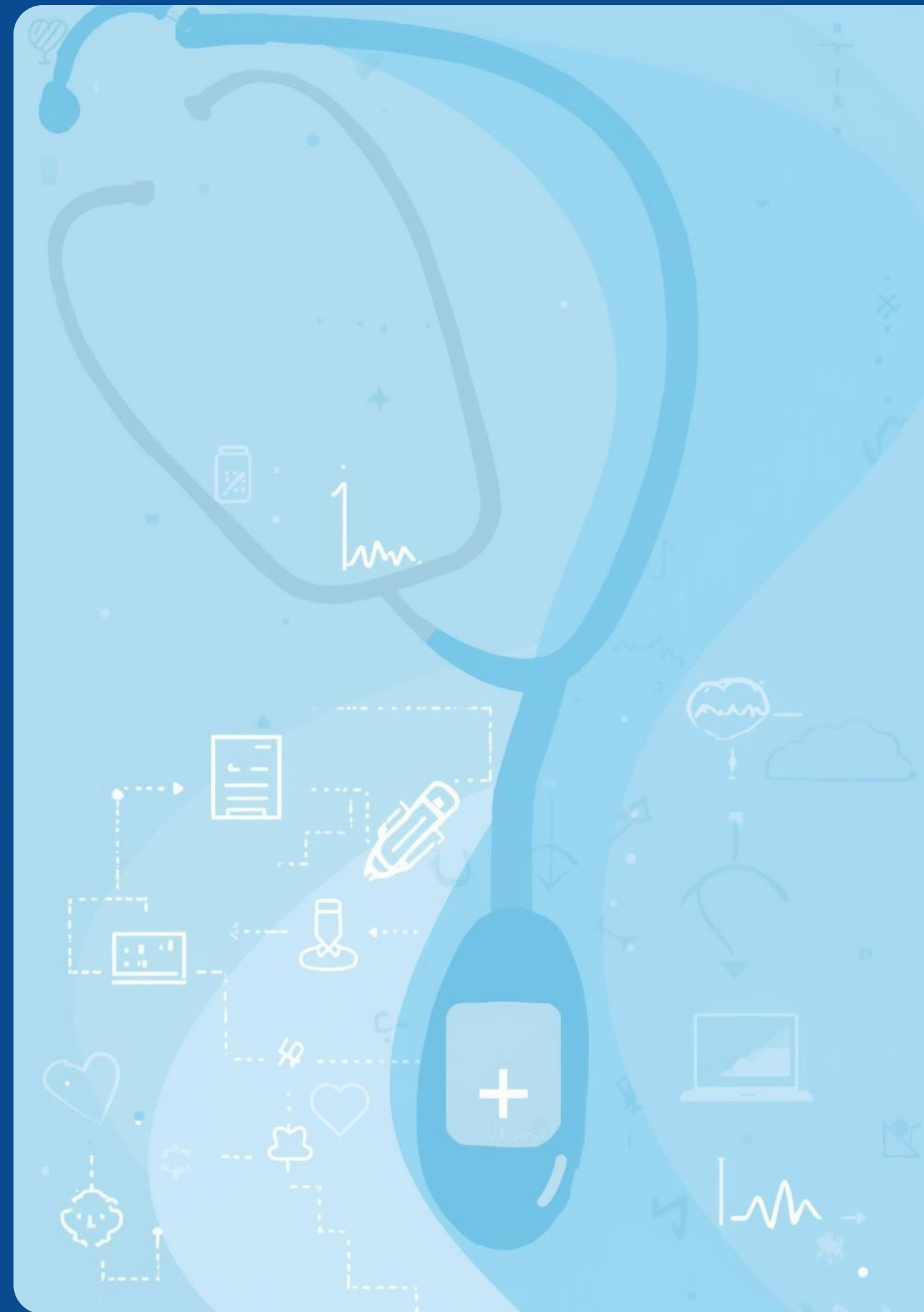


Diabetes Project

NLP Structured Extraction

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Inspiration: Bringing Dr. Lawrence Weed's Vision Into the Modern World

- Weed's Problem-Oriented Medical Record (**POMR**)
- Dr. Lawrence Weed: introduced **POMR** and **SOAP** notes
- **Vision:** medical records must be organized and computable
- Our project revives and extends Weed's idea using modern NLP:
 - Tokenization
 - ClinicalBERT Named Entity Recognition
 - Relation Extraction (labs -- values -- dates)
 - Normalization to standard vocabularies
 - Automated timeline generation
- **Goal:** Transform free-text diabetes notes into structured, interoperable data for decision support, research, and longitudinal tracking

Background: Why Diabetes Notes Are Hard to Use



Diabetes care generates large volumes of unstructured clinical text

Contain critical information not captured in structured EHR fields



Unstructured notes are difficult to search, summarize, and integrate

Synonyms, abbreviations, temporal ambiguity, and negation.



A critical need for automated NLP-driven structuring of clinical notes

Extract entities, temporal relations, and standardized concepts

Research Gaps Identified *Limitations in Existing Diabetes NLP Systems*

1. Relation Extraction (RE) still weak

Systems detect “A1c” and “9.6%” but struggle to **link test-value-unit-date**.

2. Temporal Reasoning

Notes contain vague expressions: “last year”, “recent labs”, “in 2023”.

3. Vocabulary Normalization

Inconsistent coding: Many synonyms + brand names
Needs mapping to LOINC, SNOMED, RxNorm.

4. Generalizability

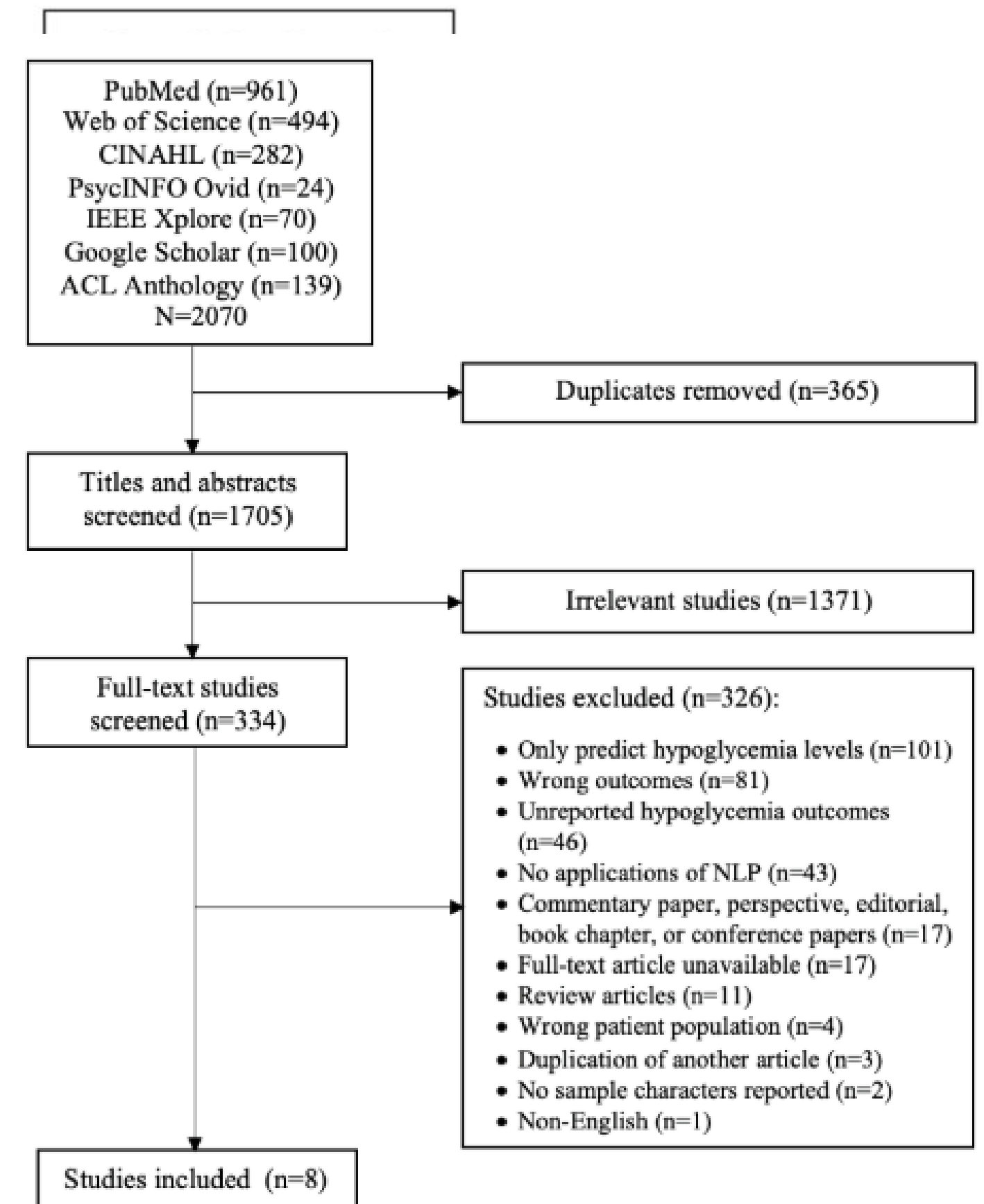
Models trained on one hospital fail in others.

5. Limited CDSS Integration

Most NLP pipelines stop at extraction: **not integrated into decision support**.

Foundational Work in Biomedical NLP

- The Role of Natural Language Processing during the COVID-19 Pandemic: Health Applications, Opportunities, and Challenges (Al-Garadi et al., 2022).
- Identifying Diabetes Related-Complications in a Real-World Free-Text EMRs Using NLP (Saban et al., 2025).
- Reconstructed HbA1c history from past notes: track glycemic control (Colmenares-Mejía et al., 2025).
- Identified hypoglycemia events using NLP (Zheng et al., 2022).
 - Classic systems (MedLEE, MetaMap, cTAKES): rule-based entity extraction.
 - Modern models (BioBERT, ClinicalBERT): better context understanding, negation, and relation modeling.



How This Project Fills the Gaps

- A complete **diabetes-focused NLP pipeline** (ClinicalBERT + rule-based hybrid)
 - **Labs, values, units, medications, complications**
 - **Temporal normalization**
 - Interoperability: **LOINC, SNOMED CT, UMLS**
- Evaluate: **accuracy, precision, recall, F1**, and visualize NER performance
- Produce structured output
 - Supports **population-wide research**

Dataset Description: Diabetes Clinical Notes (MIMIC-IV)

MIMIC-IV Diabetes Clinical Notes (Primary Dataset)

- 18,618 clinical notes
- ~3,000 unique patients
- Filtered for diabetes-related terms (A1c, insulin, glucose, metformin, neuropathy, etc.)
- Used for:
 - Real-world diabetes text
 - Cleaning, normalization, segmentation
 - Final pipeline demonstration
- MIMIC dataset was unlabeled so it provided realistic diabetes narrative text

Dataset Description Kaggle i2b2 Medication Relation Dataset (Labeled Training Data)

36,348 annotated text snippets

- Includes tags: [SUBJ]...[/SUBJ], [OBJ]...[/OBJ]

Contains **8 relation labels**:

- Reason-Drug, Route-Drug, Strength-Drug, Frequency-Drug
- Dosage-Drug, Duration-Drug, Form-Drug, ADE-Drug

Used for:

- **Training** ClinicalBERT for NER + Relation Extraction
- **Validation** and **testing** model accuracy

Split used in project (due to Colab RAM limits):

- Train: **5,000**
- Val: **1,000**
- Test: **5,453**

Pipeline Overview

1. Data Acquisition

- Reviewed diabetes-related clinical notes from MIMIC-IV for context
- Used **Kaggle Clinical Relation Extraction dataset (36,348 labeled samples)** for model training

2. Preprocessing

- Extracted entities and inserted [SUBJ]...[/SUBJ] and [OBJ]...[/OBJ] markers
- Parsed JSON labels → created unified text and relation columns
- Cleaned rows and encoded 8 relation types into numeric IDs

3. Dataset Preparation

- Split into Train (70%), Validation (15%), Test (15%)
- Tokenized using **Bio_ClinicalBERT tokenizer** and converted to HuggingFace Dataset format

4. Model Fine-Tuning

- Fine-tuned **Bio_ClinicalBERT** on medication relation extraction
- Small-RAM setup: 1 epoch, LR=2e-5, batch size=16

5. Evaluation

- Achieved high accuracy (≈94% Precision, Recall, F1)
- Generated confusion matrix, loss curve, and performance metrics

6. Inference & Application

- Built prediction function for real clinical text
- Extracts relations like Drug–Dose, Drug–Frequency, Drug–Route, Drug–Reason, and ADE
- Supports diabetes medication analysis and structured clinical summarization

Model Training + Evaluation

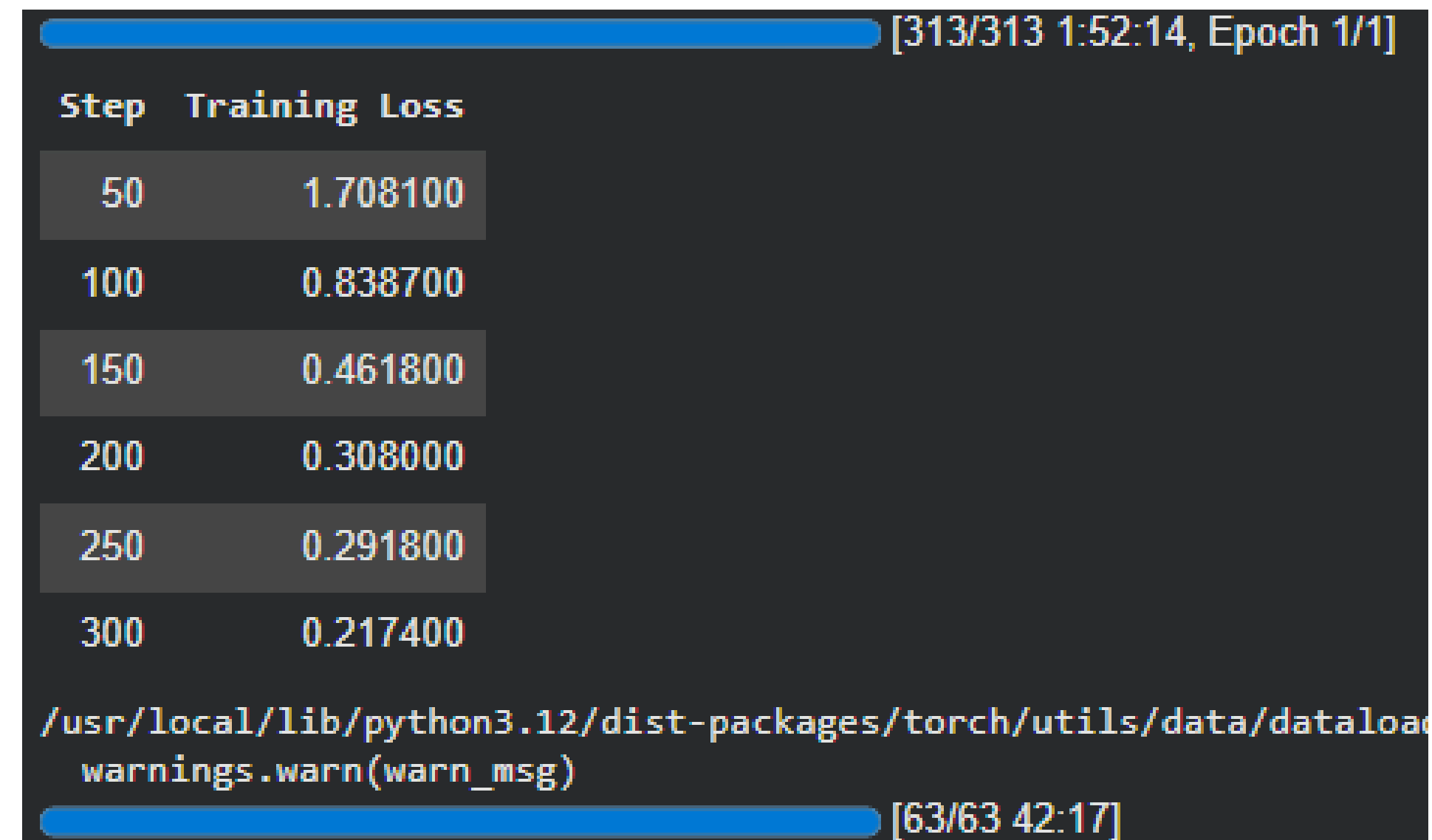
Model: Bio_ClinicalBERT (fine-tuned for clinical relation extraction)

Training Setup

- Training dataset: **25443 (70%)**
- Validation dataset: **5452 (15%)**
- Test Dataset: **5452 (15%)**
- Epochs: **1** (CPU-only runtime ~2 hours)
- Optimizer: **AdamW**
- Learning rate: **2e-5**

Training Progress

- The loss steadily decreased, showing successful learning as shown
- The smooth drop in loss shows model is learning relations correctly and there is no overfitting or divergence observed



Step	Training Loss
50	1.708100
100	0.838700
150	0.461800
200	0.308000
250	0.291800
300	0.217400

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader
warnings.warn(warn_msg)

Results: Classification Report and Train/Validation Split

```
# =====  
# 3. Train / val / test splits + label IDs  
# =====  
from sklearn.model_selection import train_test_split  
  
unique_labels = sorted(df_kaggle["relation"].unique())  
label2id = {lab: i for i, lab in enumerate(unique_labels)}  
id2label = {i: lab for lab, i in label2id.items()}  
print("Label mapping:", label2id)  
  
df_kaggle["label_id"] = df_kaggle["relation"].map(label2id)  
  
train_df, temp_df = train_test_split(  
    df_kaggle,  
    test_size=0.3,  
    random_state=42,  
    stratify=df_kaggle["label_id"]  
)  
val_df, test_df = train_test_split(  
    temp_df,  
    test_size=0.5,  
    random_state=42,  
    stratify=temp_df["label_id"]  
)  
  
print("Train:", train_df.shape, "Val:", val_df.shape, "Test:", test_df.shape)
```

Train/Validation Dataset Split

- Split dataset: Train/Validation/Test 70:15:15
- Training 25443; Validation 5452; Test 5453

	accuracy	precision	recall	f1
validation	0.943	0.944	0.943	0.941
test	0.944	0.945	0.944	0.941

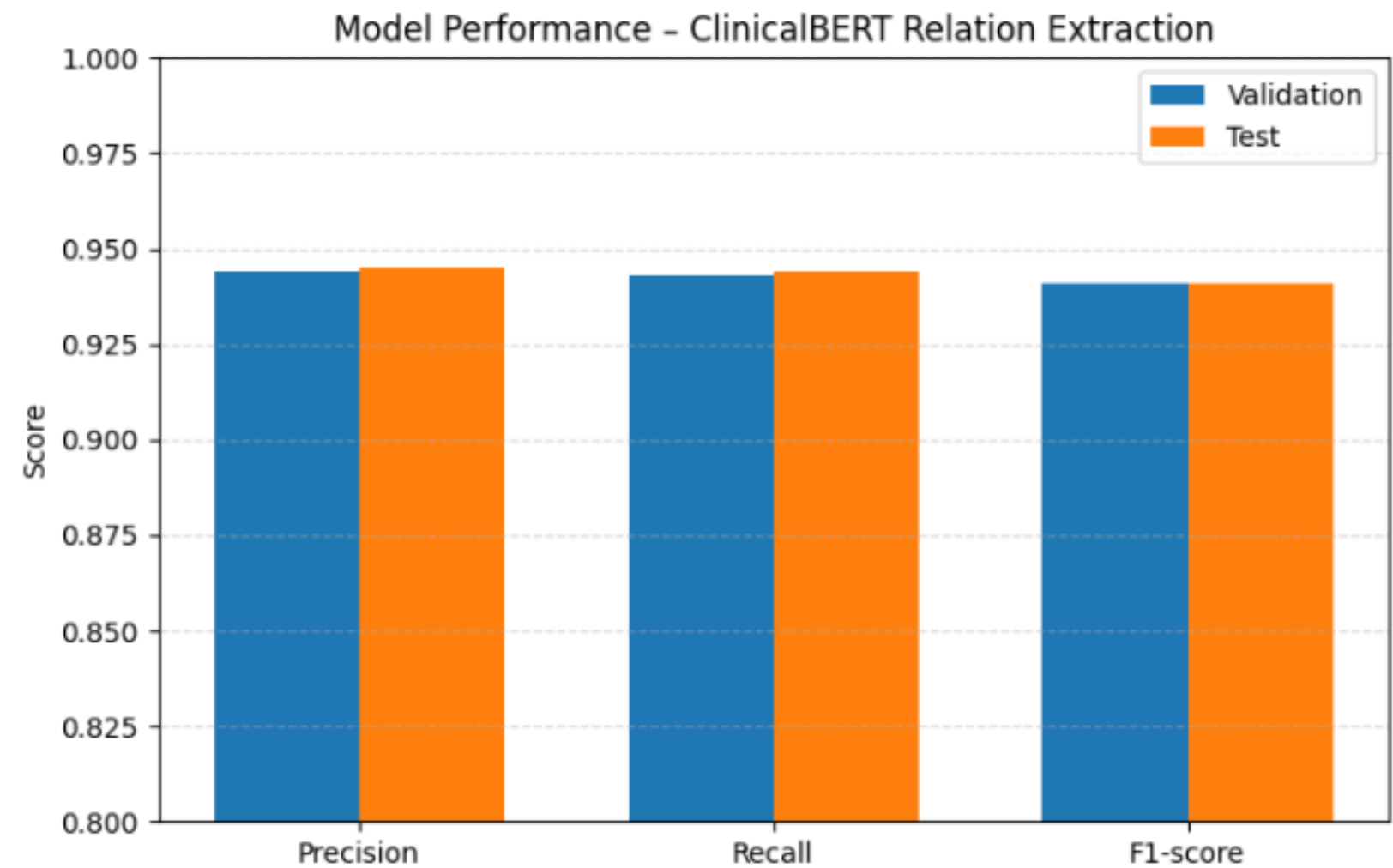
ClinicalBERT NER Performance Metrics

- High performance: all over 0.9
- Weak supervision combined with ClinicalBERT produced highly reliable extraction of diabetes-related entities

Results:Classification Report

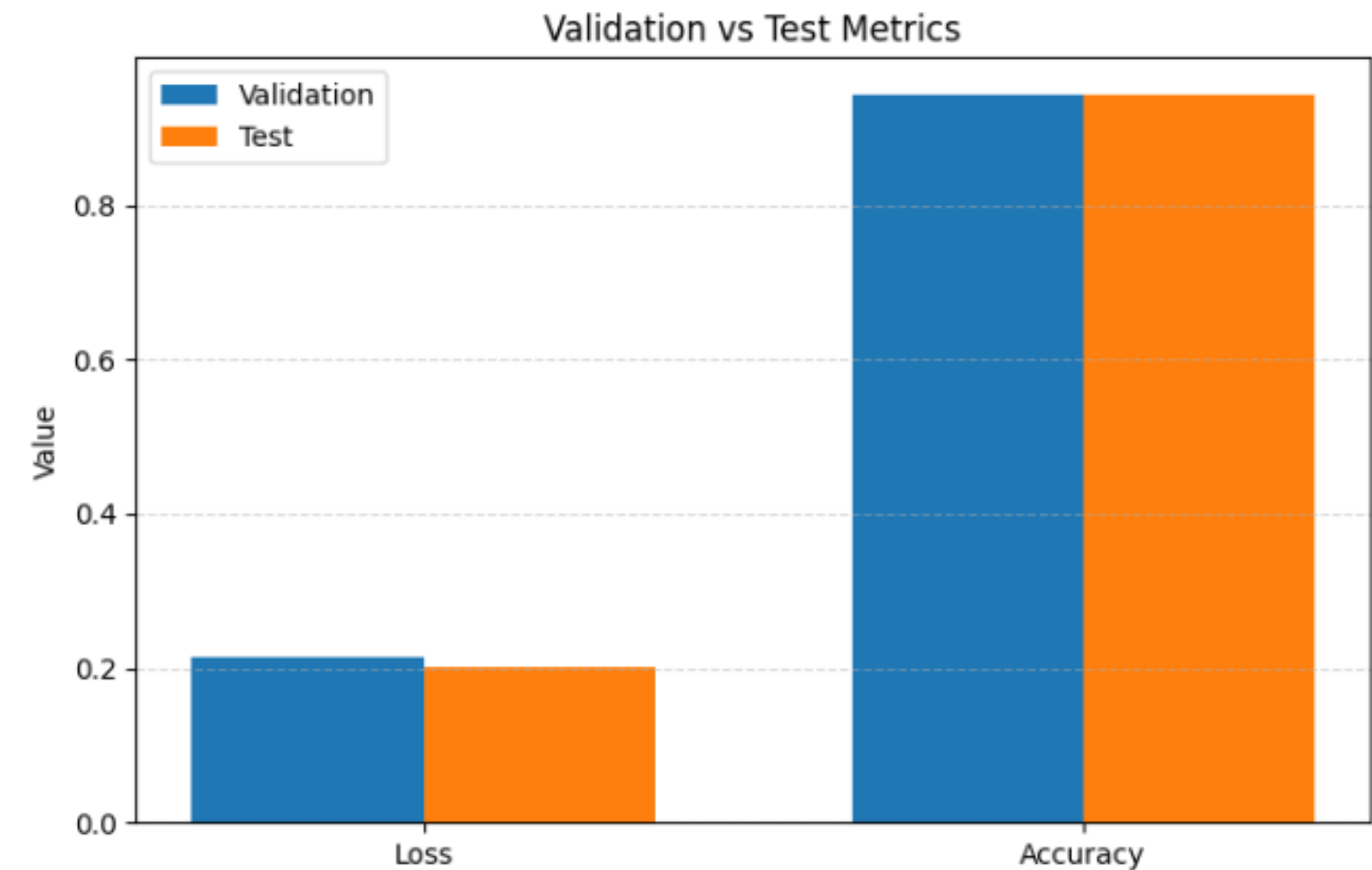
Model Performance - ClinicalBERT Extraction

High performance(0.93–0.95)
ClinicalBERT correctly identifies diabetes-relevant information
Strong generalization of weak-supervised labels after fine-tuning



Validation vs Test Metrics

Compare validation & test (loss)
Compare validation & test (accuracy)

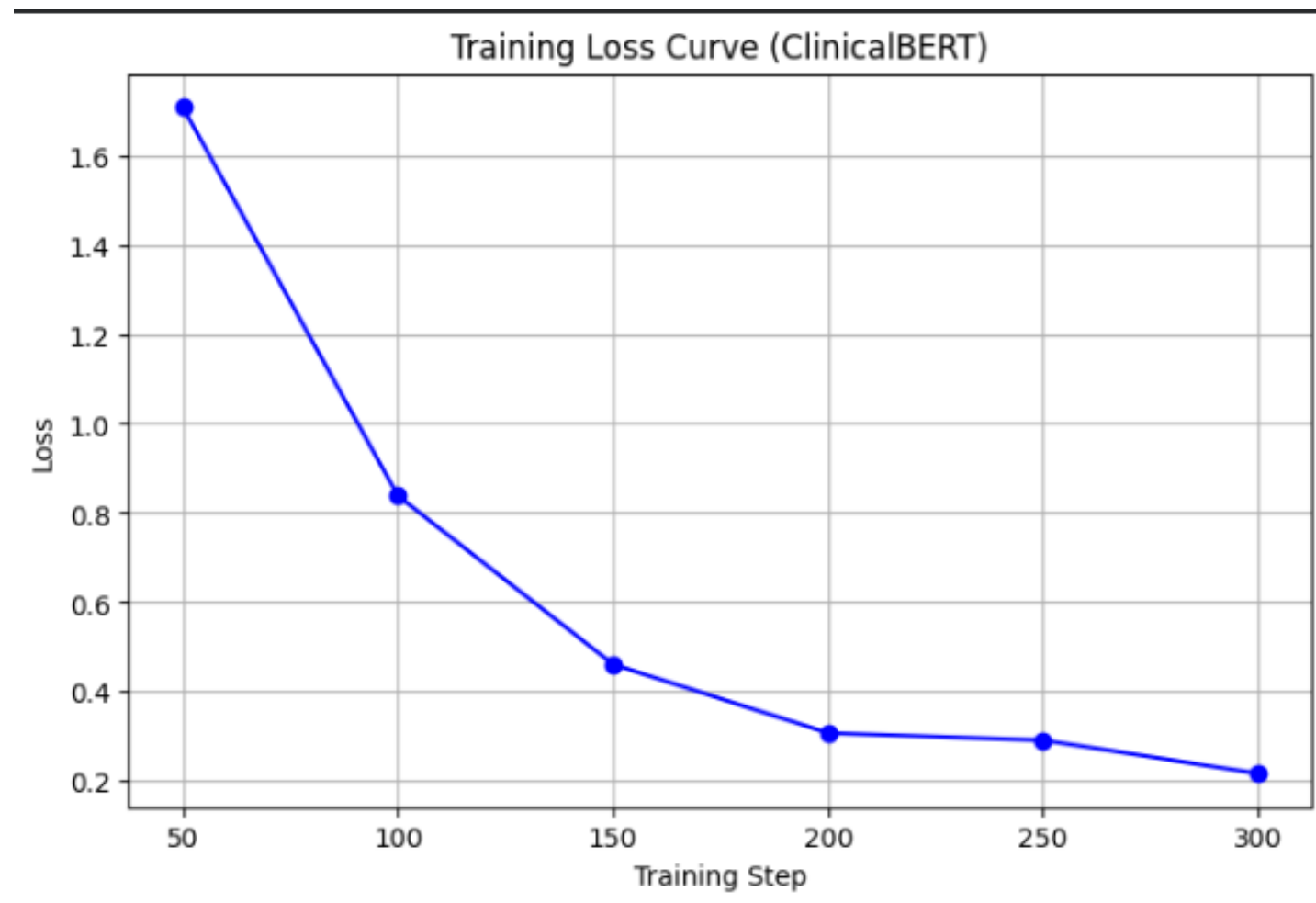


Results:Classification Report

Training Loss Curve - ClinicalBERT Extraction

Training increases, loss decreases

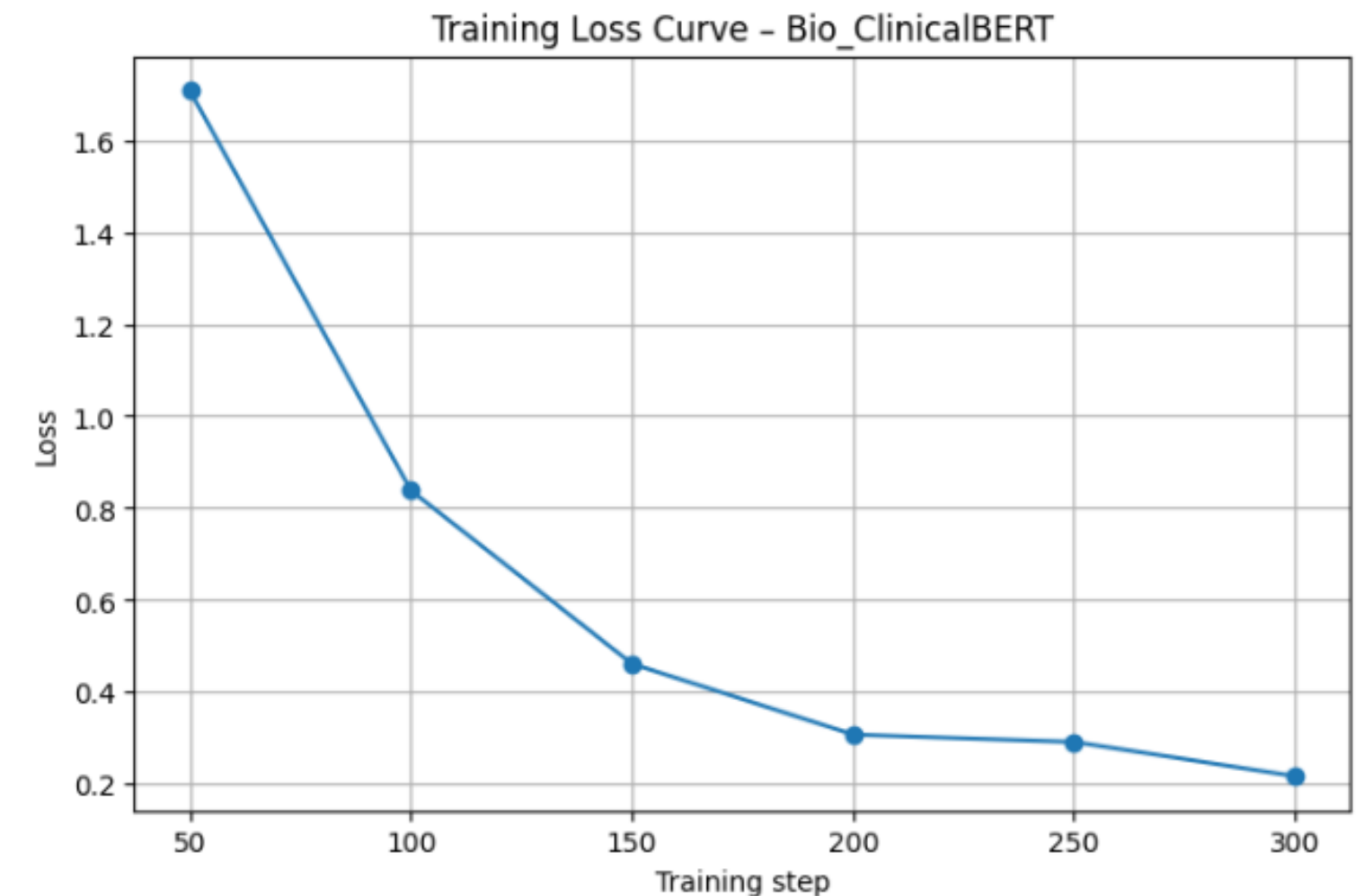
Training: 50, Loss: 1.7



Training Loss Curve - Bio_ClinicalBERT

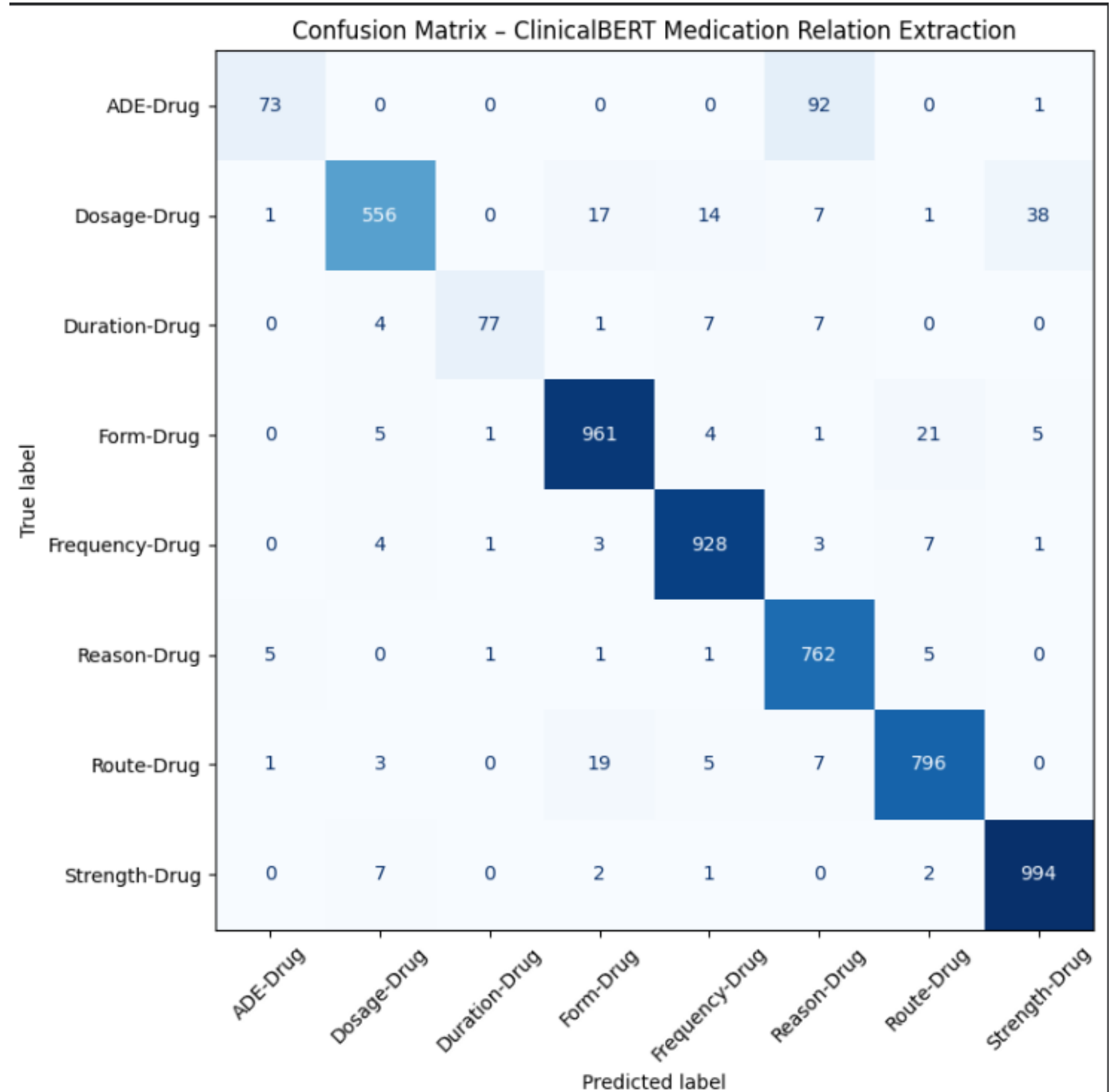
Training increases, loss decreases

Training: 50, Loss:1.7



Confusion Matrix

- ClinicalBERT Medication Relation Extraction:
 - Matrix shows a dominant diagonal (almost all predictions match true labels)
 - Very few misclassifications, even between similar token types
 - Confirms robust label differentiation in clinical language



Conclusions

- Transforms unstructured diabetic clinical notes into structured data
- The model's reliability makes it suitable for:
 - Clinical Decision Support
 - Automated chart review
 - Longitudinal diabetes tracking (A1c timelines, medication timelines)
 - Disease progression models
- Weak supervision + ClinicalBERT is a powerful and practical approach
- **Future work:** add relation extraction (e.g., A1c - value - date), improve temporal normalization, and integrate into a hospital-ready CDS pipeline

Clinical Decision Support Impact

Timelines

data extraction

timely interventions

alert systems



Early Detection

Structured data

early detection

monitoring systems.



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